



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Dr Sukanya Ledalla
Student Name	Billakanti Siri
Roll Number	2520030601
Branch	CSE
Course Outcome	CO4- Dijkstra with Edge Penalties

Case Study Topic: Hyderabad Metro — Shortest Time with Line-Transfer Penalty using Dijkstra Algorithm

Work Done Summary:

In this case study, we implemented Dijkstra's Algorithm on a Hyderabad Metro network to compute the shortest effective travel time between metro stations while considering line-transfer penalties. The metro system contains multiple lines and interchange stations. Graph transformation using virtual nodes was used to model line transfers and calculate optimal travel routes efficiently.

Implementation Details:

1. Created metro graph structure using Java adjacency lists.
2. Represented metro stations and metro lines using virtual nodes.
3. Added weighted edges representing travel time between stations.
4. Added transfer edges with 5-minute penalty at interchange stations.
5. Implemented Dijkstra's shortest path algorithm using Priority Queue.

Result: Screenshots/output

```
===== HYDERABAD METRO =====

Shortest Effective Time:
LBN -> HTC = 41 minutes
siribillakanti@siris-MacBook-Air VSC DSA II %
```

GitHub Link:

https://github.com/siribillakanti/DSA2_ASSIGNMENTS/blob/main/assignment_CO4.java